MMAUG / AI & DEVOPS BOOTCAMP

Automating Azure DevOps Workflows Using Foundry Models

Promise Uche Nwachukwu
Lead DevOps Engineer

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Meet the speaker



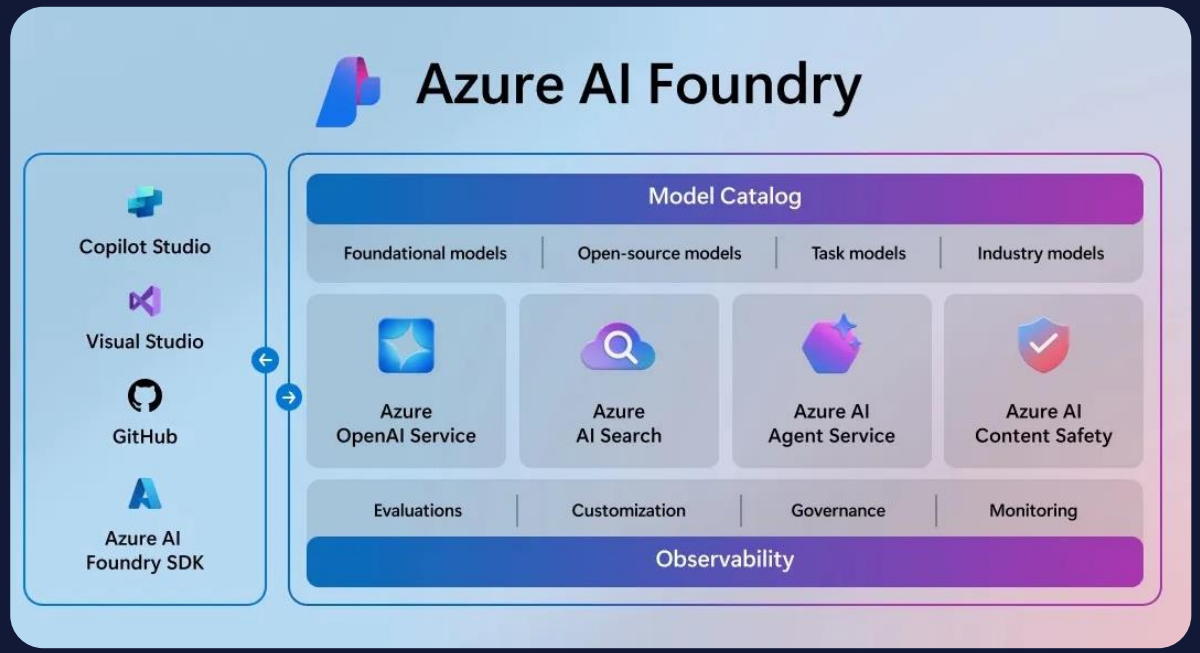
Promise Uche Nwachukwu
Lead DevOps Engineer

Introduction

- Senior Consultant and Lead DevOps Engineer with expertise in designing secure, scalable, and resilient cloud solutions.
- Proven track record in driving cloud transformation, infrastructure automation, and operational excellence across diverse industries.
- Skilled in cloud security, DevOps practices, and digital transformation strategies that align technology with business goals.

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Microsoft AI Foundry represents a comprehensive platform for building AI systems at enterprise scale



What is AI Foundry?

Microsoft AI Foundry represents a comprehensive platform **for building, deploying, and managing** intelligent **AI systems** at **enterprise scale**

It provides **the infrastructure, tools, and governance frameworks** necessary for production-grade agentic AI applications

Why Agentic AI matters now

Organizations a moving om simple chatbots to autonomous agents that can **reason, plan, and execute complex workflows**. Foundry makes this transition secure, scalable, and manageable

3X

Faster Deployment

Compared to custom solutions

99.9%

Uptime SLA

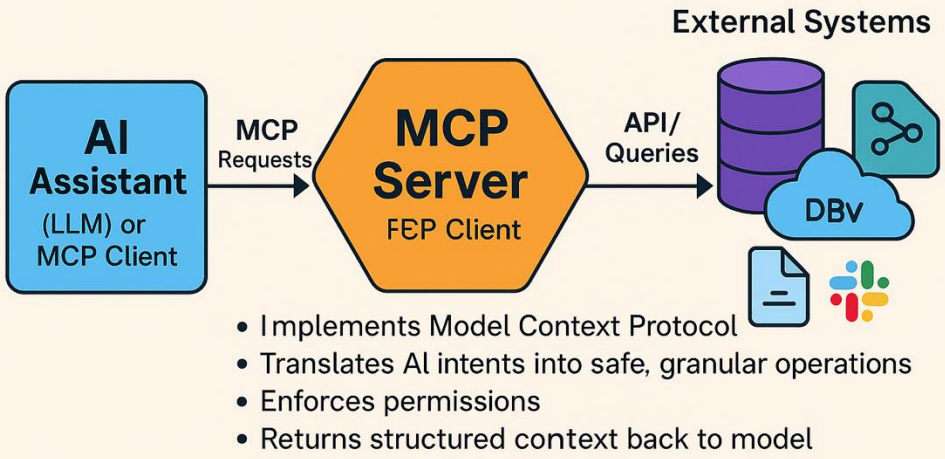
Enterprise-grade reliability



MCP Servers

MCP Servers provides standardized interfaces for agents to securely access tools, data sources and systems

What is an MCP Server?



MCP Server bridges AI ↔ Data/Tools

What they are

- **Model Context Protocol** (MCP) Servers provide **standardized interfaces** for agents to securely access tools, data sources, and external systems
- They act as **intelligent middleware** that translates agent requests into actions

Why they matter

- MCP Servers **enable consistent security policies, audit trails, and rate limiting** across all agent interactions
- They **prevent direct access** to **sensitive systems** while maintaining flexibility and performance

Key Benefits

- **Standardized security protocols**
- **Centralized access control**
- **Built-in monitoring and logging**
- **Seamless tool integration**

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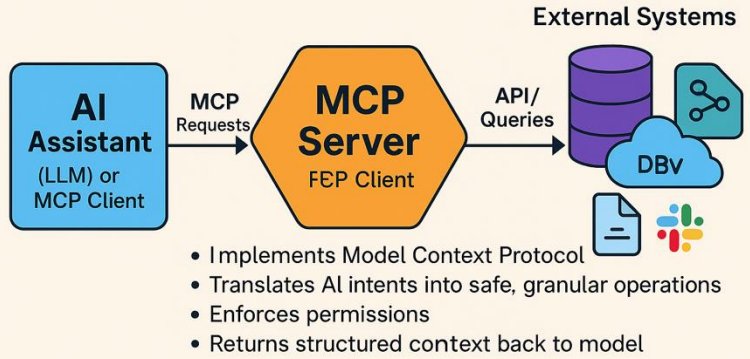
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MCP Servers can be used for enterprise teams in IT operations, Customer service, finance and healthcare

Software development & IT Operations

- ✓ Automated DevOps & CI/CD
- ✓ Infrastructure as Code (IaC)
- ✓ Real-time Observability

Customer services & Sales

- ✓ Unified Support Triage
- ✓ Sales Automation

Finance & FinOps

- ✓ Automated Cost Analysis
- ✓ Compliance & Audit

Healthcare and Smart Manufacturing

- ✓ Clinical Decision Support
- ✓ Predictive Maintenance

General business productivity

- ✓ Talent Onboarding
- ✓ Information Management

MCP Servers impact business by allowing faster time-to-market, cost reduction and eliminating integration silos



Financial & Operational impact

- ✓ Cost reduction
- ✓ Increased productivity
- ✓ Faster time-to-market

Strategic & Competitive advantages

- ✓ Vendor Independence
- ✓ Eliminating integration silos
- ✓ Future-proofing

Key business risks

- ✓ Security vulnerabilities
- ✓ Data privacy
- ✓ Implementation overhead

Industry-specific transformations

- ✓ Financial services
- ✓ Healthcare
- ✓ Manufacturing
- ✓ Business Intelligence

How to set up your first MCP server

LIVE DEMO



Configure server instance

- Define authentication, endpoints, and connection parameters in Azure portal
- Set up resource groups and networking policies for secure communication channels

Connect tools & data sources

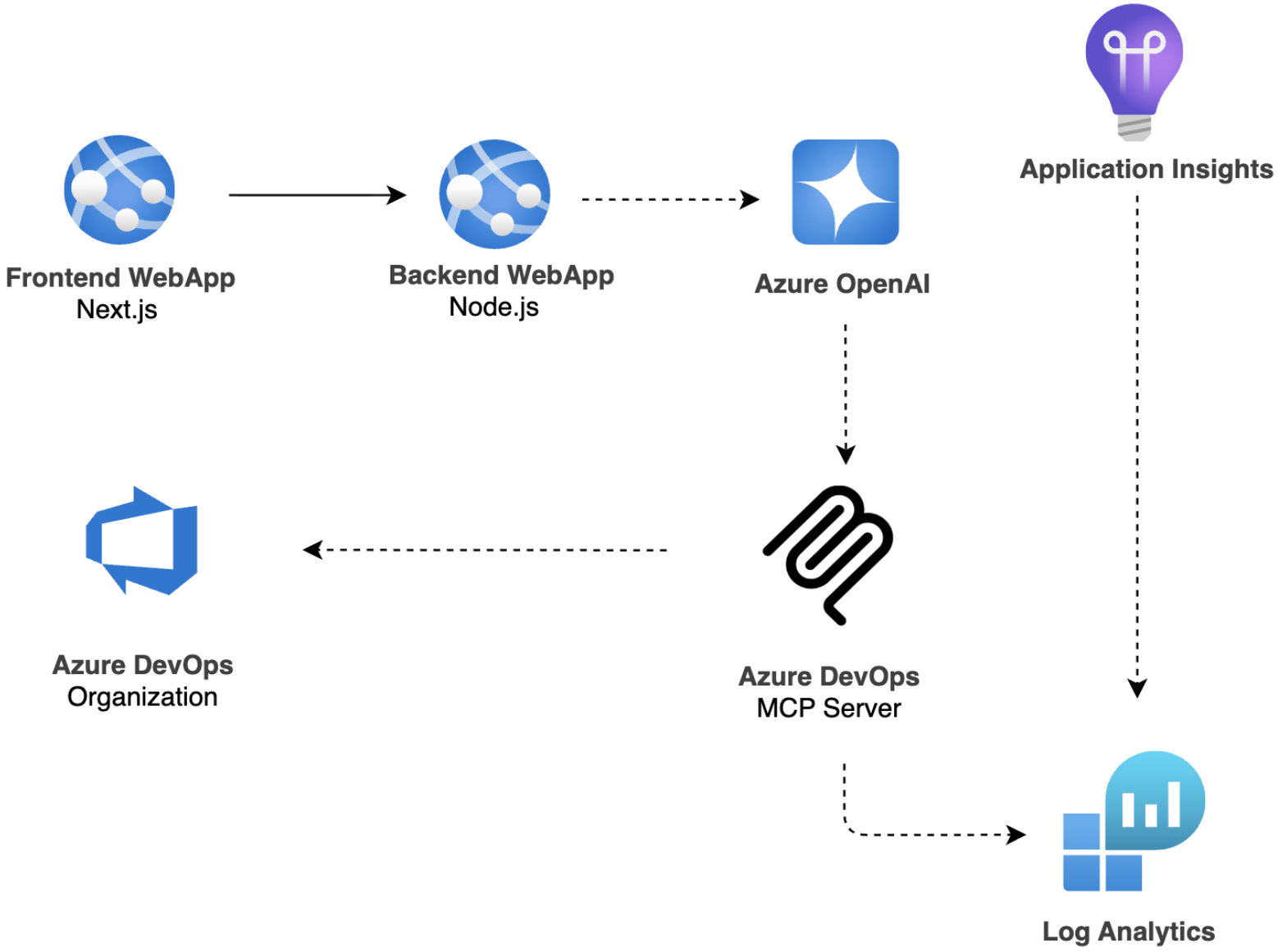
- Register available tools, APIs, and databases that agents can access
- Configure permissions and rate limits for each resource to ensure controlled access

Test & validate connection

- Run diagnostic tests to verify secure connectivity
- Validate that authentication flows work correctly and monitoring systems capture all relevant telemetry data

Pro Tip:

Start with read-only permissions during initial setup, then gradually expand access as you validate security controls and monitoring effectiveness



Azure DevOps MCP Server

Azure DevOps MCP

list the available tools you have access to?

I have access to a variety of Azure DevOps functionalities including:

- Listing projects, teams, repositories, branches, pull requests, builds, and releases.
- Managing work items: create, update, assign, query, search, get revisions, links, relations, attachments, comments, and work item types.
- Handling code-related tasks: listing commits, searching code, getting file content.
- Managing wikis: listing wikis and getting wiki pages.
- Managing project configurations: listing iterations, test plans, test suites, variable groups, service endpoints, tags, boards, board columns and rows.
- Managing team and user information.
- Handling pipelines: listing build definitions, release definitions, and pipelines.
- Creating pull requests.

If you need assistance with any specific task, just let me know!

list the repositories available

Ask about Azure DevOps work items, builds, releases...

Press `Enter` to send, `Shift + Enter` for new line



DEMO TIME

Repository: https://github.com/promiseuche/azure-devops-mcp-server